250 results for “(survey OR review) AND (wildfire OR “wild fire” OR “wildland fire” OR “wild land fire” OR “forest fire” OR bushfire) AND (propagation OR spread OR behavior OR dynamics OR trajectory) AND (prediction OR forecast OR simulation OR model OR management)”

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NATIONAL FIRE BEHAVIOUR PREDICTION SYSTEM

M. Cruz | J. Gould | Environmental Science • 2009

The estimation of **fire** behaviour is an important component of any **fire management** approach, allowing the determination of the impacts of **fire** on ecosystem components and supporting **forest fire**… Expand

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Trending and emerging prospects of physics-based and ML-based wildfire spread models: a comprehensive review

Harikesh Singh | Li-Minn Ang | +4 authors | S. Srivastava

Environmental Science, Physics • Journal of Forest Research • 27 September 2024

The significant threat of **wildfires** to **forest** ecology and biodiversity, particularly in tropical and subtropical regions, underscores the necessity for advanced predictive **models** amidst shifting… Expand

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Forest management under uncertainty: the influence of management versus climate change and wildfire in the Lake Tahoe Basin, USA.

Charles J. Maxwell | R. Scheller | J. Long | P. Manley

Environmental Science • Ecology and Society • 2022

. Climate change will accelerate **forest** mortality due to insects, disease, and **wildfire**. As a result, substantial resources will be necessary where and when **forest** managers seek to maintain multiple… Expand

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Evaluating the performance of WRF in simulating winds and surface meteorology during a Southern California wildfire event

Mukesh Kumar B. Kosović Hara P. Nayak William C. Porter J. Randerson T. Banerjee

Environmental Science · Frontiers in Earth Science · 19 January 2024

The intensity and frequency of **wildfires** in California (CA) have increased in recent years, causing significant damage to human health and property. In October 2007, a number of small **fire** events,… Expand

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Modeling of Wildfire Digital Twin: Research Progress in Detection, Simulation, and Prediction Techniques

Yuting Huang Jianwei Li Huiru Zheng

Environmental Science, Computer Science · Fire · 12 November 2024

TLDR This paper summarizes the key technologies required to establish a **wildfire** digital twin system, focusing on the technical requirements **and** research progress in **fire** detection, **simulation**, **and prediction**, **and** proposes the **wildfire** digital twin **model**, which integrates real-time data **and** computational **simulations** to replicate **and** predict **wildfire behavior**. Expand

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Facing the Wildfire Spread Risk Challenge: Where Are We Now and Where Are We Going?

Jingjing Sun Wenwen Qi Yuandong Huang Chong Xu Wentao Yang

Environmental Science · Fire · 7 June 2023

Wildfire is a sudden and highly destructive natural disaster that poses significant challenges in terms of response and rescue efforts. Influenced by factors such as climate, combustible materials,… Expand

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GIS and Machine Learning for Analysing Influencing Factors of Bushfires Using 40-Year Spatio-Temporal Bushfire Data

Wanqin He S. Shirowzhan C. Pettit

Environmental Science, Geography · ISPRS Int. J. Geo Inf. · 6 June 2022

TLDR The results of this study can inform how **fire** agencies can better understand **fire** occurrence mechanisms **and** use **bushfire prediction and simulation** techniques to support both their operational (short-term) **and** strategic **fire management** responses **and** policies. Expand

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Generative Algorithms for Fusion of Physics-Based Wildfire Spread Models with Satellite Data for Initializing Wildfire Forecasts

B. Shaddy Deep Ray +7 authors A. Oberai

Environmental Science, Physics · Artificial Intelligence for the Earth Systems · 5 September 2023

TLDR This work develops a physics-informed approach for inferring the history of a **wildfire** from satellite measurements, providing the necessary information to initialize coupled atmosphere-**wildfire models** from a measured **wildfire** state. Expand

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Using Satellite-Derived Fire Arrival Times for Coupled Wildfire-Air Quality Simulations at Regional Scales of the 2020 California Wildfire Season

W. Lassman J. Mirocha +6 authors M. Barbato

Environmental Science · Journal of Geophysical Research: Atmospheres · 10 January 2023

Wildfire frequency has increased in the Western US over recent decades, driven by climate change and a legacy of **forest management** practices. Consequently, human structures, health, and life are… Expand

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Comparison of Meteorological Drivers of Two Large Coastal Slope-Land Wildfire Events in Croatia and South-East Australia

Ivana Čavlina Tomašević V. Vučetić +4 authors B. Malečić

Environmental Science · Atmosphere · 26 June 2023

Understanding the relationship between **fire behavior** and the driving weather conditions is critical for **fire management** and long-term **fire** risk assessment. In this study, we focus on two **wildfire**… Expand

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